

Exploratory Data Analysis (EDA) on Titanic Dataset

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Objective:

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis aims to understand the dataset, identify patterns, discover relationships between variables, detect outliers, and derive meaningful insights through statistical summaries and visualizations.

Dataset Overview

The Titanic dataset contains information about passengers aboard the Titanic, including demographic details, ticket information, passenger class, fare paid, and survival status.

Dataset Statistics:

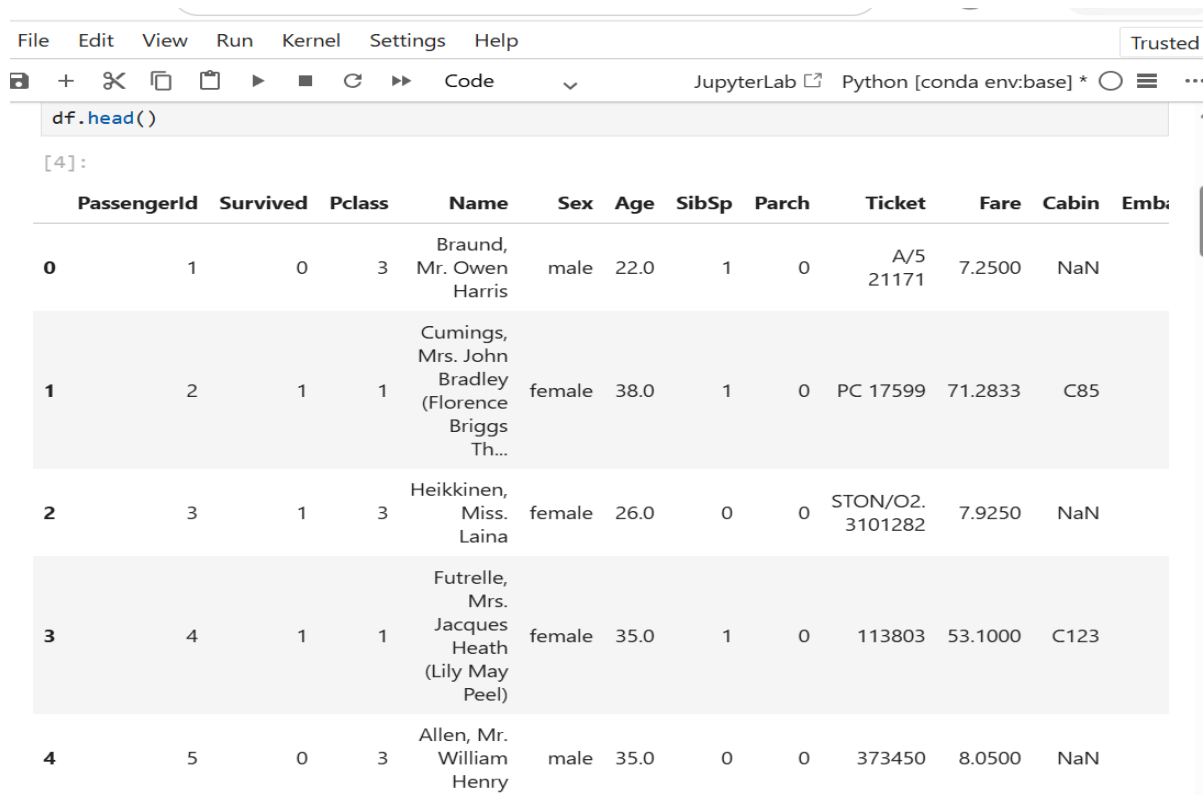
- Total Rows: 891
- Total Columns: 12

Missing Values:

- Age: 177 missing values
- Cabin: 687 missing values
- Embarked: 2 missing values

Screenshots:

1. df.head()



```
df.head()
```

[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	

2. df.info()

```
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df.info

[6]:
<bound method DataFrame.info of      PassengerId  Survived  Pclass  \
0               1         0        3
1               2         1        1
2               3         1        3
3               4         1        1
4               5         0        3
..          ...     ...     ...
886           887         0        2
887           888         1        1
888           889         0        3
889           890         1        1
890           891         0        3

      Name      Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2  Heikkinen, Miss. Laina    female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0      1
4  Allen, Mr. William Henry    male  35.0      0
..          ...     ...     ...
886  Montvila, Rev. Juozas    male  27.0      0
887  Graham, Miss. Margaret Edith    female  19.0      0
888  Johnston, Miss. Catherine Helen "Carrie"    female   NaN      1
889  Behr, Mr. Karl Howell    male  26.0      0
890  Dooley, Mr. Patrick    male  32.0      0

      Parch      Ticket    Fare Cabin Embarked
```

3. df.describe()

```
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[7]:
df.describe()

[7]:
      PassengerId  Survived  Pclass     Age  SibSp  Parch    Fare
count  891.000000  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008    0.381594   32.204208
std    257.353842    0.486592    0.836071   14.526497    1.102743    0.806057   49.693429
min      1.000000    0.000000    1.000000    0.420000    0.000000    0.000000    0.000000
25%    223.500000    0.000000    2.000000   20.125000    0.000000    0.000000    7.910400
50%    446.000000    0.000000    3.000000   28.000000    0.000000    0.000000   14.454200
75%    668.500000    1.000000    3.000000   38.000000    1.000000    0.000000   31.000000
max    891.000000    1.000000    3.000000   80.000000    8.000000    6.000000  512.329200

[8]:
df.isnull().sum()

[8]:
PassengerId    0
Survived       0
```

4. `df.isnull().sum()`

```
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max 891.000000 1.000000 3.000000 80.000000 8.000000 6.000000 512.329200

[8]:
df.isnull().sum()

[8]:
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64

[10]:
```

Univariate Analysis

1. Age Distribution

Observation:

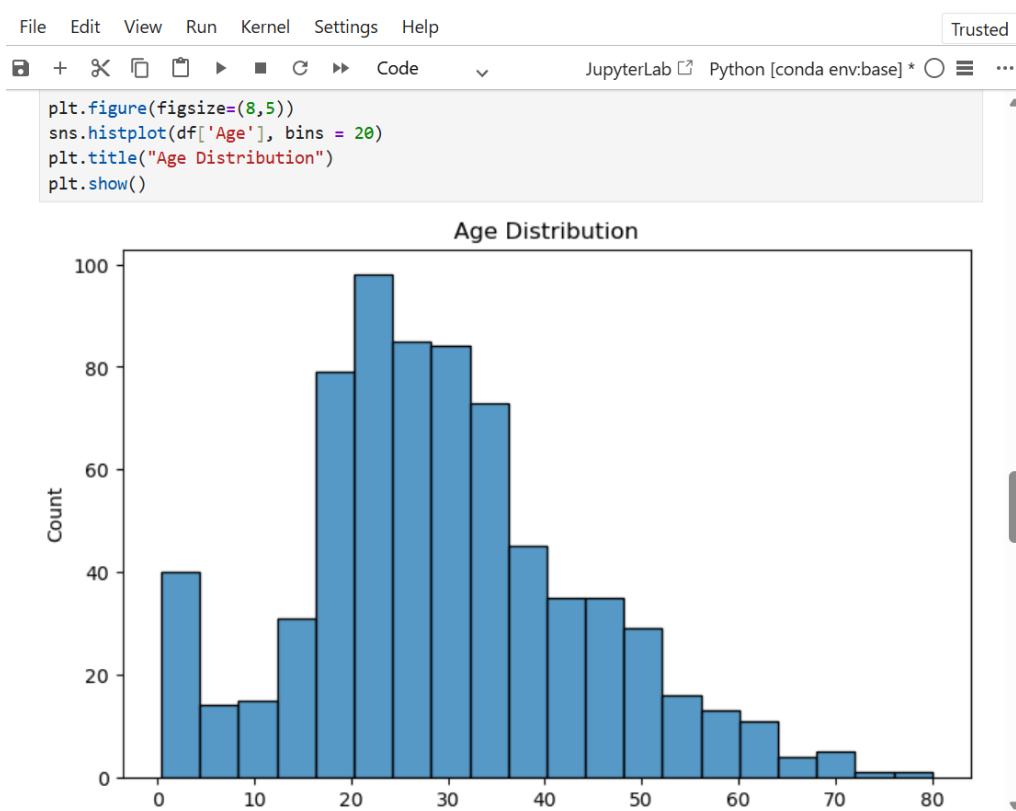
Most passengers were between 20 and 40 years old. The dataset contains passengers from infants to approximately 80 years old.

2. Fare Distribution (Boxplot)

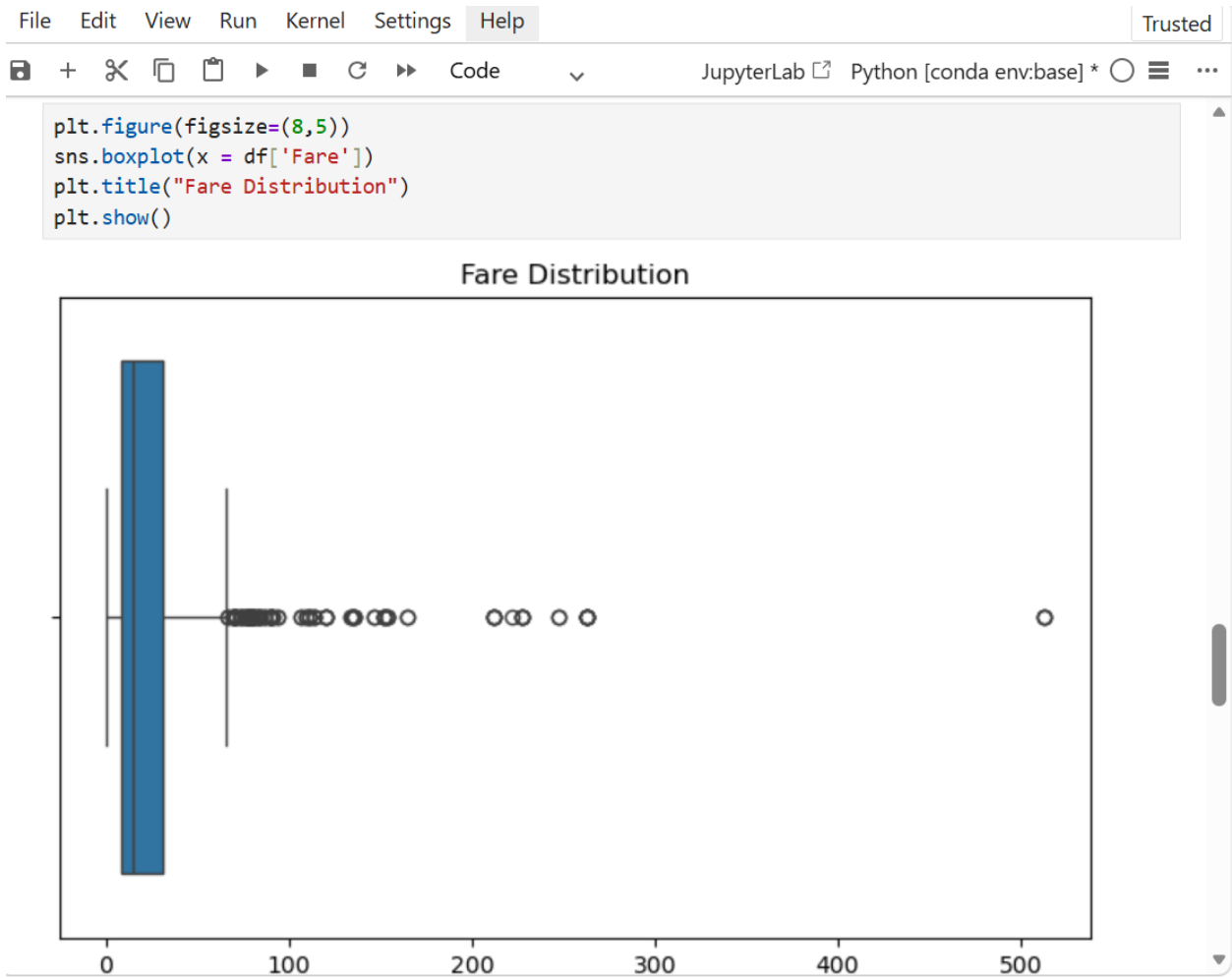
Observation:

The fare column contains several outliers. Most passengers paid relatively low fares, while a few passengers paid significantly higher fares.

1. Age Distribution



2. Fare Boxplot



Bivariate Analysis

1. Survival by Gender

Observation:

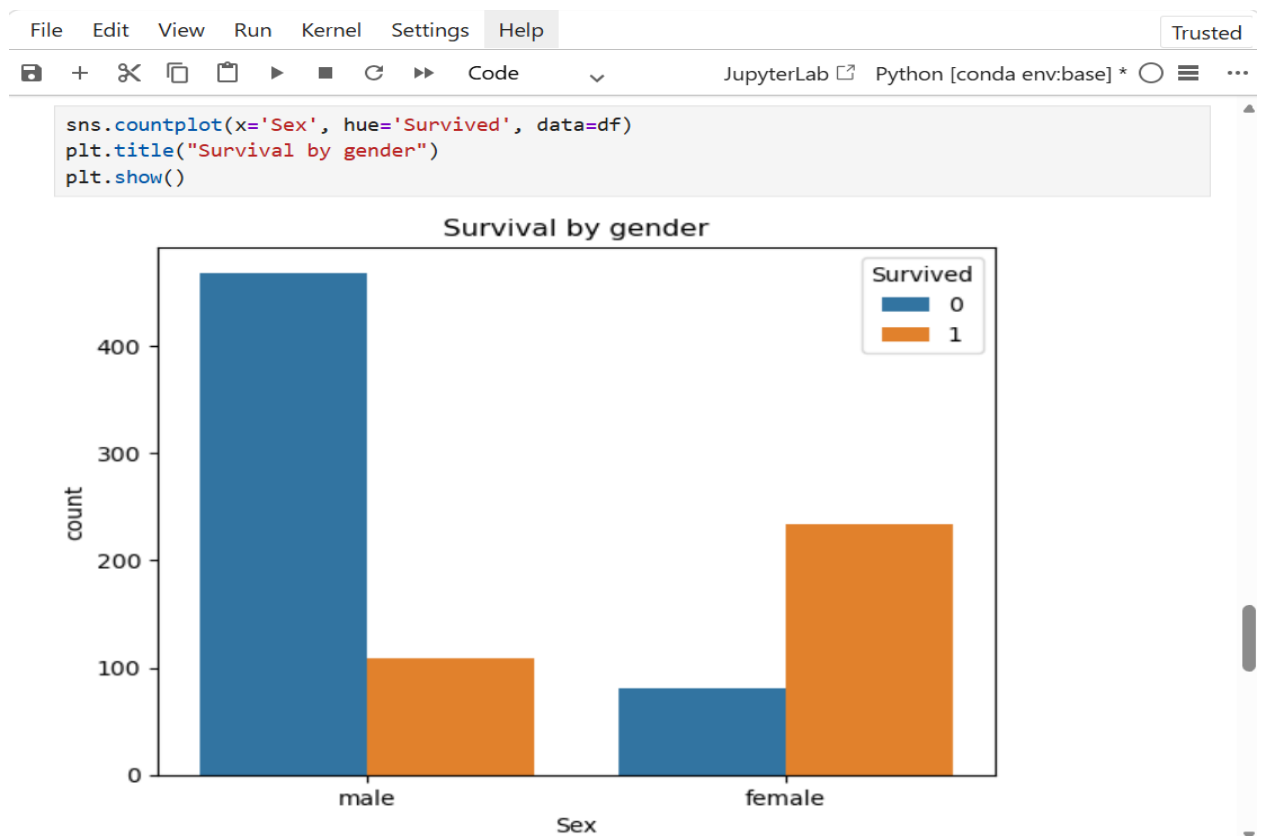
Female passengers had a significantly higher survival rate than male passengers.

2. Age vs Fare Scatter Plot

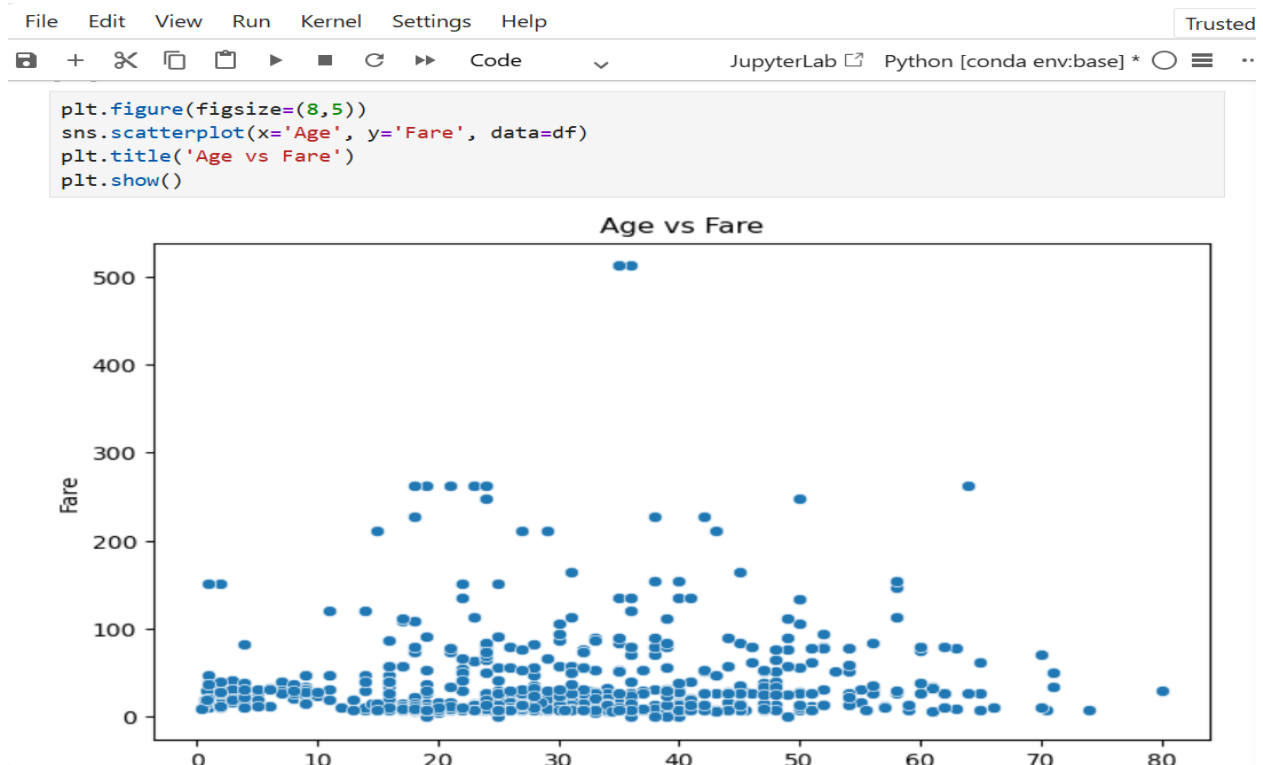
Observation:

There is no strong relationship between Age and Fare. Most passengers paid lower fares regardless of age, while a few passengers paid very high fares.

1. Survival by gender



2. Scatter Plot



Correlation Analysis

1. Correlation Heatmap

Observation:

Fare has a positive correlation with survival (0.26), while Passenger Class has a negative correlation with survival (-0.34).

2. Pairplot

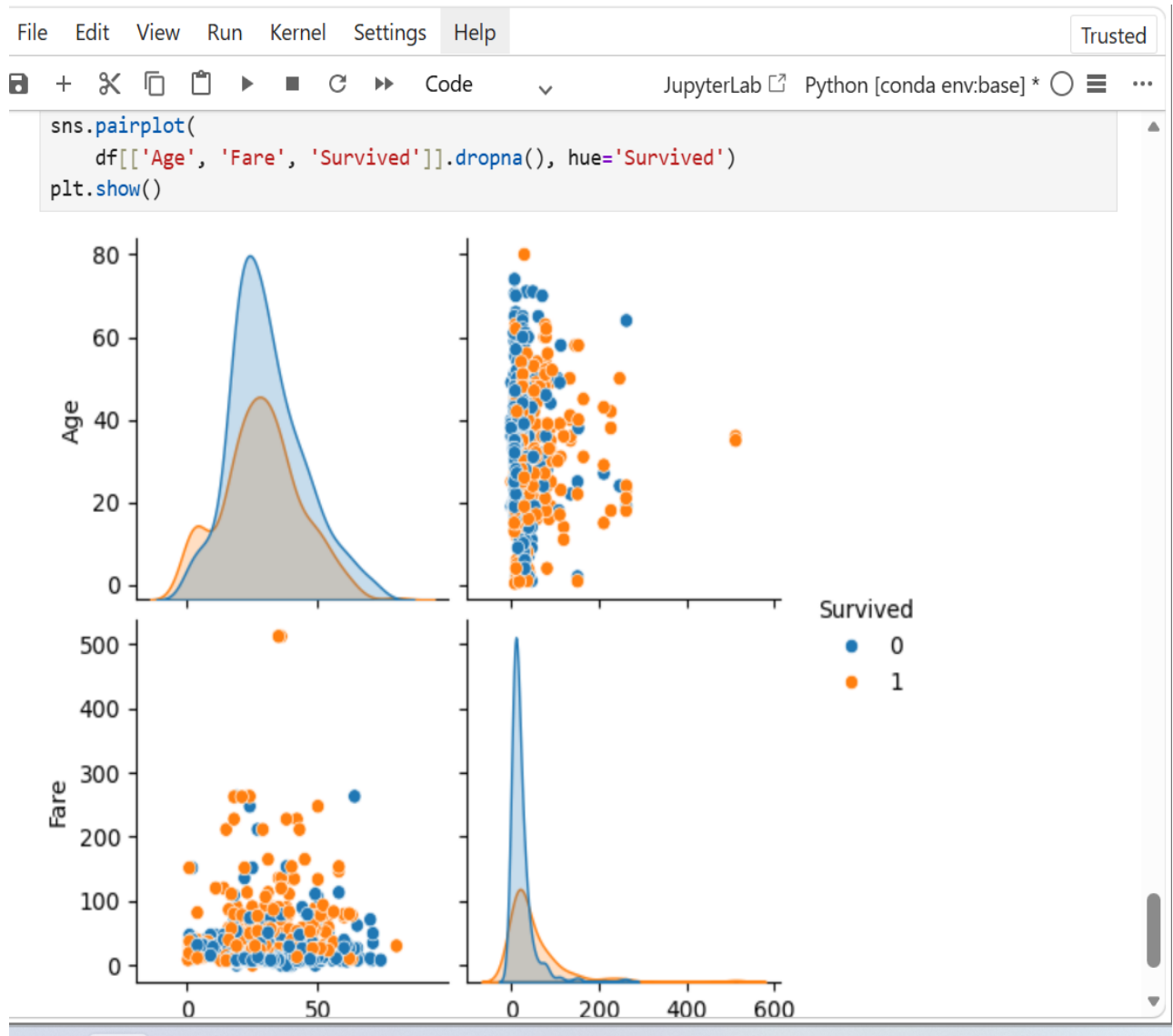
Observation:

Survivors generally tend to have higher fare values. No strong linear relationship exists between Age and Fare.

1. Heatmap



2. Pairplot



Conclusion

Key Findings:

- Female passengers had a higher survival rate than male passengers.
- Passenger class significantly influenced survival.
- Most passengers were between 20 and 40 years old.
- Fare distribution contains several outliers.
- Higher fare-paying passengers had better chances of survival.
- Missing values were mainly present in Age and Cabin columns.

The Exploratory Data Analysis successfully identified important patterns, relationships, and trends within the Titanic dataset. These insights can support future predictive modeling and machine learning tasks.